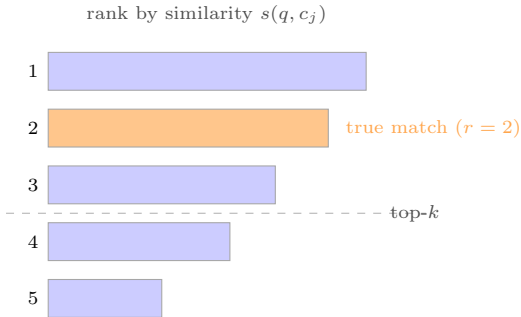




$\text{recall}@k = \mathbf{1}\{r(q) \leq k\}$; $\text{MRR} = 1/r(q)$. Here $r = 2$: a $\text{recall}@1$ miss, $\text{recall}@3$ hit, reciprocal rank $1/2$.